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LEARNER GUIDE

For

ISO 14064-1:2018 Greenhouse Gas (GHG) Emissions and Removals

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Conducted by

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	Legacy release	Legacy trainer slide deck.	Tertiary Infotech Academy
2.0	20 August 2026	Complete source-aligned visual rebuild; expanded ISO 14064-1 inventory, risk, reduction, renewable-energy and offset content; ten connected activities and aligned assessments.	Dr Alfred Ang

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How to Use This Guide

The slide deck is concept-led. This guide carries the detailed procedures, scenarios, questions, evidence requirements and acceptance checks for all ten activities. Each activity also has a self-contained folder under activities/ with a working Excel workbook and printable PDF/checklist.

Use only fictional, anonymised or authorised workplace information. Apply confidentiality, due professional care, transparent calculation and evidence-based judgement throughout.

Course Learning Outcomes

- LO1: Innovate organisational GHG accounting systems using current climate science, ISO 14064-1 principles and broader sustainability initiatives.
- LO2: Develop risk-informed GHG management plans that meet relevant reporting and regulatory requirements and support mitigation decisions.
- LO3: Develop operational emissions-reduction strategies across product lifecycles, value chains, supply chains and renewable-energy options.
- LO4: Generate economically viable carbon-offset and carbon-credit strategies aligned with organisational targets and residual-emissions needs.

Competency Alignment

Code	Competency requirement
K1	Evolving research on GHG dynamics and innovation in GHG accounting
K2	Integration of GHG accounting with sustainability and climate initiatives
K3	GHG reduction across product lifecycles, value chains and supply chains
K4	Renewable-energy technologies, economics, sustainability impacts and regulation
K5	Best-practice approaches for GHG accounting and reporting
K6	Relevant regulations, programmes and reporting guidelines
K7	Frameworks for evaluating, mitigating and monitoring GHG-related risks
K8	Approaches and economics of carbon-offset projects
K9	Sustainability frameworks integrated into GHG-reduction strategy
A1	Develop reduction strategies across product lifecycles, value chains and supply chains

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Code	Competency requirement
A2	Generate the economic costs and benefits of offset projects to determine suitability
A3	Innovate organisational GHG accounting using emerging science and accounting practice
A4	Develop GHG management frameworks, plans and monitoring initiatives
A5	Develop a carbon strategy for project selection and credit optimisation
A6	Influence economically viable renewable-energy strategies aligned with sustainability goals
A7	Develop GHG accounting and reporting guidelines integrated with sustainability initiatives
A8	Define assessments and mitigation strategies for GHG-related risks

Current Standards Baseline

- ISO 14064-1:2018 specifies organisation-level principles and requirements for designing, developing, managing and reporting GHG inventories.
- ISO 14064-2 addresses project-level emission reductions and removal enhancements; ISO 14064-3 addresses validation and verification of GHG statements.
- The 2018 revision uses a significance-based approach to indirect emissions and organises reporting into direct emissions plus five indirect-emission categories.
- Confirm the authorised standard text, programme rules, legal obligations, factor/GWP sources and reporting-period requirements before a formal conformance or public claim.

Topic 01 — Foundations of Organisational GHG Accounting Systems

Alignment: K1 · K2 · K5 · A3 · A7 · LO1

GHG science · ISO 14064 family · accounting principles · targets · sustainability integration

GHG science for organisational decisions

- **Radiative forcing:** GHGs absorb and re-emit infrared energy, changing the Earth's energy balance.
- **Long-lived stocks:** Carbon dioxide accumulates; current emissions add to a persistent atmospheric stock.

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- **Different gases:** Methane, nitrous oxide and fluorinated gases differ in lifetime and warming effect.
- **CO2e:** A common unit converts gas mass using a declared global-warming-potential basis.

Auditor application

- Inventory all relevant gases
- State the GWP source and assessment basis
- Use science to improve priorities
- Separate physical fact from target choice

Judgement cautions

- CO2e is a comparison convention
- A lower annual total does not erase historic stock
- GWP values can change by assessment
- Avoid false precision beyond the data

ISO 14064 family and assurance architecture

- **Part 1:** Organisation-level principles and requirements for inventory design, quantification and reporting.
- **Part 2:** Project-level quantification, monitoring and reporting of emission reductions or removal enhancements.
- **Part 3:** Principles and requirements for verification and validation of GHG statements.
- **Related frameworks:** GHG Protocol, ISO 14067 and programme rules can complement—but not silently replace—the chosen criteria.

Auditor application

- Name the inventory criteria
- Define the intended assurance use
- Keep organisation and project claims distinct
- Resolve conflicting programme rules

Judgement cautions

- Certification is not automatic
- A project credit is not an organisational removal
- One framework mapping may not be exact
- Use authorised standards text for formal conformance

ISO 14064-1 principles in practice

- **Relevance:** Information and boundaries must serve intended users and decisions.
- **Completeness:** Include significant sources and justify omissions transparently.

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- **Consistency:** Methods and boundaries support comparison through time.
- **Accuracy and transparency:** Reduce bias and uncertainty and disclose assumptions, evidence and limitations.

Auditor application

- Turn each principle into a review question
- Use materiality and significance deliberately
- Document changes at the time they occur
- Prefer traceable evidence over unsupported detail

Judgement cautions

- Principles interact and require judgement
- Completeness does not mean perfect data
- Consistency does not freeze weak methods
- Transparency does not excuse avoidable error

Organisational and reporting boundaries

- **Purpose first:** Intended users and decisions determine the inventory design.
- **Consolidation:** Equity share, financial control or operational control changes which operations are included.
- **Reporting boundary:** Direct and significant indirect emissions are organised into decision-useful categories.
- **Boundary evidence:** Ownership, leases, contracts, operating authority and influence support the decision.

Auditor application

- Map all entities before excluding any
- Apply one consolidation approach consistently
- Document outsourced activities
- Reassess after acquisitions or divestments

Judgement cautions

- Legal ownership and operational control can differ
- Outsourcing does not make emissions disappear
- Boundary choice can change trends
- Avoid cherry-picking low-emission operations

Science-aligned targets and target adjustments

- **Target boundary:** Specify gases, operations, categories and coverage so ambition can be compared with the inventory.
- **Base year:** Choose a representative period and publish the recalculation policy before structural change occurs.

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- **Pathway:** Connect near-term milestones and initiatives to an explicit trajectory rather than a distant headline alone.
- **Adjustment:** Recalculate for material structural or methodology changes; do not rewrite history for ordinary performance variance.

Auditor application

- Link target scope to inventory scope
- Define a significance threshold
- Separate recalculation from restatement
- Retain approvals and an adjustment bridge

Judgement cautions

- A science-aligned target is not an inventory requirement
- Growth assumptions must remain visible
- Acquisitions can break comparability
- Offsets do not repair an inconsistent base year

Quantification, base year and report integrity

- **Activity data:** Measured or estimated quantities describe the emission-causing activity.
- **Emission factor:** A documented coefficient translates activity into gas mass or CO2e.
- **Base year:** A stable comparison point needs recalculation rules for structural and methodological change.
- **Report:** Disclose boundaries, methods, exclusions, factors, uncertainty, changes and performance separately.

Auditor application

- Retain a calculation trail
- Reconcile to primary records
- Recalculate under a written policy
- Report gross inventory before offsets

Judgement cautions

- Units cause frequent large errors
- Spend data can mask physical change
- Estimated data must be labelled
- Credits must not be netted into gross emissions

Activity 1 — Set the GHG Inventory Purpose and Boundaries

Alignment: K1 · K2 · K5 · A3 · LO1 · Duration: 60 minutes

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Objective: Define intended users, reporting boundaries and consolidation choices for a credible organisational GHG inventory.

Scenario

EcoForge Asia Pte Ltd (EFA) manufactures precision cooling equipment in Tuas. It owns 80% of the Tuas plant, holds 40% of a Batam component venture, leases a Singapore office, contracts a Johor warehouse and uses third-party distribution. The board wants a 2025 inventory suitable for management decisions and future independent verification, while the bank wants evidence supporting a sustainability-linked loan.

Materials

- Group-structure and facility fact sheet
- List of intended users and decisions
- Operational-control and equity-share prompts
- Boundary diagram template

Activity Questions

- Who will use the inventory and what decisions must it support?
- Which entities and facilities are included under the selected consolidation approach?
- Which exclusions would be material or misleading?
- What evidence will justify each boundary decision?

Detailed Procedure

1. Read the EcoForge fact pattern and distinguish verified facts from assumptions or missing information.
2. Write a concise inventory purpose covering intended users, reporting period, decisions and desired assurance readiness.
3. Map each entity, facility, lease and outsourced operation, including ownership and operational-control facts.
4. Compare operational control, financial control and equity share; select one approach and document why it fits the purpose.
5. Apply the chosen approach consistently to each entity and record included, excluded and unresolved items.
6. Define the reporting boundary for direct emissions, imported energy, transportation, products, use of products and other indirect sources relevant to EcoForge.
7. Identify one removal or reservoir question, including whether onsite trees or purchased credits belong in the organisational inventory.
8. Challenge exclusions against relevance, completeness, transparency and decision-usefulness.
9. Peer-review another team's boundary and resolve one disagreement with evidence.

Required Evidence

- Inventory purpose statement
- Organisational boundary diagram
- Consolidation decision log

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- Peer-review record

Acceptance Check

The diagram covers every named entity and operation; the consolidation method is applied consistently; every exclusion and unresolved item has a transparent rationale and owner.

Activity files: activities/01-/README.md, workbook and printable PDF/checklist.*

Activity 2 — Build a Significant GHG Source and Sink Register

Alignment: K1 · K5 · A3 · A7 · LO1 · Duration: 60 minutes

Objective: Classify direct and indirect GHG sources, sinks and reservoirs and assess their significance for reporting.

Scenario

EcoForge uses natural gas ovens, company vans, refrigerants, purchased electricity and rooftop solar. It buys aluminium and electronics, imports components, outsources warehousing, sells energy-consuming equipment, sends scrap for recycling and has staff travel and commuting. A landscaping contractor claims the site trees remove carbon, but no quantified evidence exists.

Materials

- Operational process narrative
- Energy, fleet and procurement summaries
- ISO 14064-1 category prompts
- Significance scoring worksheet

Activity Questions

- Which sources are direct and which are indirect by category?
- Which GHGs and removal claims require investigation?
- What significance criteria are defensible for EcoForge?
- Who owns each source record and what primary evidence exists?

Detailed Procedure

10. Walk through the value chain from purchased materials to product end of life and list every plausible source, sink and reservoir.
11. Classify sources into direct emissions, imported energy, transportation, products used, use of products and other indirect emissions.
12. Identify relevant gases such as carbon dioxide, methane, nitrous oxide and refrigerants without assuming carbon dioxide alone.
13. Define significance criteria using magnitude, influence, risk, stakeholder interest, data quality and reduction opportunity.
14. Score each indirect source and record the evidence or judgement supporting the score.

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15. Select significant indirect sources for inclusion and document transparent reasons for any exclusion.
16. Assign a data owner, data source, frequency and evidence type to each included source.
17. Assess the landscaping removal claim and state what quantification and permanence evidence would be needed.
18. Run a completeness challenge using utility bills, ledger accounts, facility walk-through and procurement categories.

Required Evidence

- Completed source/sink register
- Significance criteria and scores
- Inclusion/exclusion log
- Data-owner matrix

Acceptance Check

The register contains all scenario sources, uses the six-category logic consistently, treats removals separately from offsets and gives evidence-based significance decisions.

Activity files: activities/02-/README.md, workbook and printable PDF/checklist.*

Activity 3 — Quantify a Base-Year Inventory and Reporting Guide

Alignment: K1 · K2 · K5 · A3 · A7 · LO1 · Duration: 75 minutes

Objective: Apply activity data, emission factors, GWP values, uncertainty judgement and base-year recalculation rules to produce a transparent inventory.

Scenario

EcoForge has electricity, gas, diesel, refrigerant top-up, purchased aluminium, inbound freight and business-travel data. Units are inconsistent, one electricity month is estimated, the refrigerant factor source is unclear and an acquired line will enter the organisation in 2026. Management wants 2025 as the base year and a rule for future structural and methodological changes.

Materials

- Activity-data pack
- Illustrative emission-factor and GWP register
- Calculation and uncertainty workbook
- Base-year policy template

Activity Questions

- What conversions and unit controls are required?
- Which factor vintage and GWP basis will be disclosed?
- How should uncertainty and estimated data affect decisions?
- What events trigger base-year recalculation?

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Detailed Procedure

19. Validate source documents, reporting period, units and completeness before calculating.
20. Convert all activity data into factor-compatible units and retain the conversion trail.
21. Apply the equation activity data multiplied by emission factor multiplied by any required GWP conversion.
22. Keep biogenic carbon dioxide, removals and purchased offsets separately identified from inventory emissions.
23. Reconcile totals to source documents and investigate the missing electricity month and refrigerant-factor provenance.
24. Rate uncertainty qualitatively for activity data, factors, models and boundary decisions; propose proportionate improvements.
25. Draft a base-year recalculation policy for acquisitions, divestments, methodology changes and significant error corrections.
26. Draft reporting rules covering boundaries, methods, factor sources, significant exclusions, uncertainty and changes.
27. Perform an independent peer recalculation of two rows and resolve differences.

Required Evidence

- Quantified inventory workbook
- Factor and GWP register
- Uncertainty improvement log
- Base-year and reporting guideline

Acceptance Check

Every result traces to source data, units, factor and GWP basis; totals reconcile; uncertainty and recalculation triggers are explicit; no offset is netted into the inventory.

Activity files: activities/03-/README.md, workbook and printable PDF/checklist.*

Topic 02 — Regulatory-Compliant and Risk-Responsive GHG Management

Alignment: K6 · K7 · A4 · A8 · LO2

Reporting requirements · inventory review · governance · climate risk · control and conformance

Reporting requirements and conformance

- **Applicable obligations:** Laws, schemes, contracts and lender or customer terms can create different requirements.
- **Reporting criteria:** The inventory must name the standard, programme rules and reporting period used.

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- **Mandatory disclosures:** Boundaries, categories, methods, exclusions, uncertainty and changes need controlled evidence.
- **Approval:** Management accepts responsibility for the GHG statement before external use.

Auditor application

- Maintain an obligation register
- Translate each requirement into evidence
- Track submission and approval dates
- Control public claim versions

Judgement cautions

- Voluntary does not mean consequence-free
- Different users may need different views
- Legal applicability is context-specific
- Do not imply verification before it occurs

Inventory data governance and internal control

- **Ownership:** Every source has an accountable owner and competent preparer.
- **Method control:** Factors, GWP values, calculations and estimates are versioned and approved.
- **Evidence control:** Source records are complete, retrievable, protected and retained.
- **Review control:** Independent checks, reconciliations and issue closure support a reliable statement.

Auditor application

- Use a source-level RACI
- Lock formulas and controlled inputs
- Log changes and approvals
- Review high-risk sources first

Judgement cautions

- Sustainability should not own all data
- A spreadsheet can hide silent changes
- Review is more than proofreading
- One approval cannot compensate for weak evidence

Reviewing a GHG inventory report

- **Boundary test:** Confirm included entities, categories and significant exclusions match documented decisions.
- **Calculation test:** Reperform samples from source record through unit conversion, factor and total.

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- **Disclosure test:** Check claims, estimates, limitations and changes are described accurately.
- **Trend test:** Challenge whether structural change, method change or data gaps explain apparent performance.

Auditor application

- Trace both top-down and bottom-up
- Sample anomalies and high magnitude
- Resolve contradictions
- Retest every correction

Judgement cautions

- A balanced total can contain offsetting errors
- A factor source without vintage is weak
- Narrative claims require the same control as numbers
- Verification readiness is not verification

GHG risk landscape

- **Inventory risk:** Incomplete sources, wrong units, weak factors, estimates or uncontrolled formulas distort results.
- **Transition risk:** Policy, carbon cost, technology and market expectations can affect competitiveness.
- **Physical risk:** Acute and chronic hazards can disrupt operations, data and value-chain performance.
- **Claims risk:** Ambiguous boundaries or unsupported neutrality and reduction statements damage trust.

Auditor application

- Write cause-event-consequence statements
- Assess current-control effectiveness
- Use thresholds and early indicators
- Link material risks to management review

Judgement cautions

- Risk is not only the reported number
- High uncertainty can be decision material
- Physical and transition risks can interact
- Greenwashing risk is a governance issue

Risk-responsive GHG management plan

- **Objectives:** Connect inventory reliability and reduction outcomes to strategy and obligations.

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- **Controls:** Prevent, detect and correct data, method, performance and claim failures.
- **Monitoring:** Use leading and lagging indicators with thresholds and escalation.
- **Learning:** Review changes, incidents, findings and stakeholder challenges to improve the system.

Auditor application

- Prioritise by consequence and control weakness
- Assign decision rights
- Test response scenarios
- Report residual risk transparently

Judgement cautions

- A register without action is not management
- Too many KPIs obscure warning signals
- Risk acceptance needs authority
- Plans must evolve with business change

Activity 4 — Review a GHG Inventory and Resolve Quality Findings

Alignment: K5 · K6 · A4 · LO2 · Duration: 75 minutes

Objective: Evaluate inventory evidence, controls, conformance and report completeness before management approval or verification.

Scenario

EcoForge's draft report states a 12% reduction but the base year was not recalculated after acquiring a production line. A refrigerant leak was recorded in maintenance notes but omitted from the inventory, two factor sources are undated, a freight estimate is described as measured data and the report nets purchased credits against gross emissions.

Materials

- Draft GHG report
- Calculation and evidence extracts
- ISO 14064-1 reporting checklist
- Review and finding workbook

Activity Questions

- Which issues affect conformance, accuracy or transparency?
- What evidence is sufficient to confirm each issue?
- Which corrections must occur before approval?
- What limitations should remain disclosed?

Detailed Procedure

28. Set review objectives, scope, criteria and independence before examining the report.

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29. Trace reported totals to category subtotals, calculation rows and primary evidence samples.
30. Reperform selected calculations and confirm units, factors, GWP basis and rounding.
31. Check boundary consistency, significant indirect emissions, removals, biogenic emissions and offset presentation.
32. Compare the claimed trend with base-year recalculation policy and structural-change evidence.
33. Classify each issue as corrected error, unresolved limitation, control weakness or disclosure gap.
34. Write findings with criterion, objective evidence, evaluation and required disposition without prescribing unsupported solutions.
35. Verify corrections and prepare an approval brief listing residual uncertainty and management decisions.
36. Peer-calibrate finding severity and resolve differences in judgement.

Required Evidence

- Review plan and sample log
- Finding register
- Corrected report disclosures
- Management approval brief

Acceptance Check

All seeded defects are detected; each finding is evidence-based; corrections are rechecked; the final brief distinguishes resolved issues from residual limitations.

Activity files: activities/04-/README.md, workbook and printable PDF/checklist.*

Activity 5 — Build a GHG Governance, Compliance and Risk Plan

Alignment: K6 · K7 · A4 · A8 · LO2 · Duration: 60 minutes

Objective: Design roles, controls, compliance monitoring and mitigation actions for material GHG risks.

Scenario

EcoForge faces lender reporting covenants, customer product-footprint requests, Singapore energy and carbon-cost exposure, flood risk at the Johor warehouse, supplier transition risk and reputational risk from unsupported green claims. Responsibility is fragmented across Finance, Facilities, Procurement and Sustainability.

Materials

- Obligation and stakeholder fact sheet
- Climate-risk event cards
- Current role descriptions
- Risk-control workbook

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Activity Questions

- Which requirements are mandatory, contractual or voluntary?
- Who is accountable for inventory approval and source controls?
- Which risks need treatment, monitoring or escalation?
- What indicators provide early warning?

Detailed Procedure

37. Create an obligations register showing source, applicability, owner, required output, frequency and evidence.
38. Map decision rights using responsible, accountable, consulted and informed roles for data, methods, review and claims.
39. Describe each GHG-related risk as cause, event and consequence rather than a vague topic.
40. Assess likelihood, consequence, velocity and current-control effectiveness using documented criteria.
41. Select mitigation actions following avoid, reduce, transfer/share and accept logic.
42. Define preventive, detective and corrective controls for high-priority inventory and transition risks.
43. Choose monitoring indicators, thresholds, owners and escalation routes.
44. Test the plan against a flood disruption and a customer challenge to an emissions claim.
45. Present a concise governance recommendation to the executive sponsor.

Required Evidence

- Compliance register
- GHG governance RACI
- Risk and control register
- Monitoring dashboard and escalation protocol

Acceptance Check

Every applicable obligation has an owner and evidence; high risks have proportionate controls and indicators; accountability and escalation are unambiguous.

Activity files: activities/05-/README.md, workbook and printable PDF/checklist.*

Topic 03 — Operationalising Emissions Reduction Across Value Chains

Alignment: K3 · K4 · A1 · A6 · LO3

Boundaries · source data · baselines · lifecycle and supplier action · renewable energy

From inventory to reduction hierarchy

- **Avoid:** Remove unnecessary activity, demand or material before optimising it.

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- **Reduce:** Improve efficiency, yield, maintenance, logistics and process control.
- **Substitute:** Change fuel, material, technology, product design or service model.
- **Neutralise residuals:** Consider high-integrity removals or offsets only for clearly defined residual emissions.

Auditor application

- Start with hotspots and influence
- Quantify expected abatement
- Check rebound and burden shift
- Separate forecast from realised result

Judgement cautions

- Largest is not always most controllable
- Low-cost action can have low impact
- Avoided emissions differ from inventory reductions
- Offsetting is not operational reduction

Lifecycle and value-chain thinking

- **Upstream:** Materials, capital goods, fuel supply, inbound logistics and supplier operations.
- **Operations:** Energy, process emissions, refrigerants, waste and owned transport.
- **Downstream:** Distribution, product use, maintenance, end of life and customer behaviour.
- **Interfaces:** Design and procurement choices can shift emissions between stages and organisations.

Auditor application

- Map physical flows before calculating
- Find decision levers at interfaces
- Use functional comparisons
- Check end-of-life assumptions

Judgement cautions

- Spend is not a lifecycle model
- Recycling claims need yield and allocation
- Efficiency can trigger rebound
- Do not double count supplier and product action

Supplier data and engagement ladder

- **Estimate:** Screen with spend, mass or industry-average factors while labelling limitations.
- **Request:** Ask for activity, facility or product data using a clear specification.

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- **Collaborate:** Improve methods, capability and reduction plans with strategic suppliers.
- **Contract and verify:** Embed criteria, evidence, improvement and assurance expectations into sourcing.

Auditor application

- Segment by emissions and influence
- Standardise data requests
- Reward data improvement
- Track both coverage and reduction

Judgement cautions

- A supplier claim may use a different boundary
- Primary data can still be poor
- Questionnaires create burden
- Changing supplier may export risk rather than reduce it

Reduction baselines, targets and monitoring

- **Baseline:** State period, boundary, method, quality and structural-change treatment.
- **Target:** Define absolute or intensity metric, coverage, ambition, milestones and responsibility.
- **Pathway:** Model business-as-usual and initiative effects without hiding growth assumptions.
- **Monitoring:** Track implementation, realised activity change, emission factors and residual gaps.

Auditor application

- Use annual and interim milestones
- Reconcile projects to inventory
- Separate activity and factor effects
- Reforecast when assumptions change

Judgement cautions

- Intensity can fall while absolute emissions rise
- Targets do not prove achievement
- Forecast abatement may overlap
- Grid decarbonisation is not an internal project

Renewable energy and economic integrity

- **Efficiency first:** Reduce demand to lower cost, capacity and exposure.
- **Onsite generation:** Model technical yield, self-consumption, export, maintenance and asset life.

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- **Electrification:** Evaluate process suitability, load, grid trajectory, resilience and capital needs.
- **Procurement:** Assess contract, certificate geography, vintage, exclusivity, retirement and claim rules.

Auditor application

- Compare NPV and cost per tCO2e
- Run tariff and yield sensitivities
- State inventory accounting treatment
- Stage investment with decision gates

Judgement cautions

- A certificate does not reduce physical onsite use
- Market-based claims require controls
- Cheap credits can distract from efficiency
- Technology choice must preserve safety and quality

Activity 6 — Diagnose Hotspots and Set a Reduction Baseline

Alignment: K3 · A1 · LO3 · Duration: 50 minutes

Objective: Use inventory, lifecycle and business data to identify hotspots, set a defensible baseline and frame reduction opportunities.

Scenario

EcoForge's largest reported sources are purchased aluminium, product electricity use, imported electricity, air freight and refrigerant leakage. Data quality varies: electricity is metered, aluminium uses spend-based data, use-phase emissions rely on assumptions and freight distances are estimated.

Materials

- 2025 category inventory
- Product lifecycle facts
- Data-quality ratings
- Hotspot and baseline workbook

Activity Questions

- Which sources are largest and which are most influenceable?
- Where can lifecycle burden shift occur?
- Which baseline metrics suit growth and efficiency decisions?
- What data improvements could change prioritisation?

Detailed Procedure

46. Rank sources by absolute emissions, trend, data quality, organisational influence and stakeholder significance.

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47. Map the product lifecycle from material extraction through manufacturing, transport, use and end of life.
48. Identify potential burden shifts, such as lightweight material that increases energy use or recycling claims that ignore yield loss.
49. Define an absolute baseline and at least one intensity metric with denominator rationale.
50. Separate near-term operational levers from supplier, product-design and customer-use levers.
51. Create an opportunity shortlist using abatement potential, feasibility, cost, lead time and evidence confidence.
52. Run a sensitivity check on the two weakest high-impact assumptions.
53. State which additional primary data is needed before investment approval.

Required Evidence

- Hotspot dashboard
- Lifecycle map
- Baseline and intensity statement
- Prioritised opportunity shortlist

Acceptance Check

Priorities reflect magnitude, influence and uncertainty; baseline and intensity denominator are explicit; no recommendation creates an undisclosed lifecycle burden shift.

Activity files: activities/06-/README.md, workbook and printable PDF/checklist.*

Activity 7 — Design a Supplier and Product Reduction Roadmap

Alignment: K3 · K9 · A1 · LO3 · Duration: 60 minutes

Objective: Develop quantified value-chain actions, supplier engagement, product levers and monitoring milestones.

Scenario

Top suppliers provide mixed data: one aluminium supplier has a verified product footprint, two provide facility averages, and several only provide spend and weight. Product engineers can redesign casing mass and fan efficiency, Logistics can change mode and load factor, and Procurement can include emissions criteria at renewal.

Materials

- Supplier segmentation data
- Product and logistics options
- Illustrative cost and abatement estimates
- Roadmap workbook

Activity Questions

- Which suppliers warrant deep engagement first?

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- How will data improve from estimates to supplier-specific evidence?
- Which product and logistics actions deliver credible reductions?
- How will progress and rebound effects be monitored?

Detailed Procedure

54. Segment suppliers by emissions, strategic importance, influence, data maturity and transition readiness.
55. Define a supplier-data ladder from spend-based estimates to verified product or supplier-specific data.
56. Design engagement routes: request, educate, collaborate, contract, incentivise and escalate.
57. Quantify product-material, efficiency, freight-mode and load-factor opportunities using transparent assumptions.
58. Check each option for quality, safety, cost, lifecycle burden shift and double counting.
59. Prioritise initiatives using abatement, feasibility, marginal cost, lead time and strategic value.
60. Create 30-, 90-, 180- and 720-day milestones with owners, dependencies and decision gates.
61. Define leading and lagging indicators, including supplier response, primary-data coverage and realised tCO2e reduction.
62. Conduct a red-team review of one claim and one forecast before roadmap approval.

Required Evidence

- Supplier segmentation and data ladder
- Quantified initiative cards
- 24-month roadmap
- KPI and governance plan

Acceptance Check

The roadmap contains quantified, owned and time-bound actions; supplier data improves progressively; product actions address lifecycle trade-offs and claim boundaries.

Activity files: activities/07-/README.md, workbook and printable PDF/checklist.*

Activity 8 — Develop a Renewable-Energy Business Case

Alignment: K4 · A6 · LO3 · Duration: 55 minutes

Objective: Compare efficiency, electrification and renewable-energy options using emissions integrity, economics and implementation risk.

Scenario

EcoForge uses 4.2 GWh of grid electricity and natural gas process heat. Its roof can host 650 kWp solar, one oven may be electrified, chilled-water optimisation promises

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savings, and suppliers offer renewable-electricity contracts and certificates. Capital is constrained and the lender requires transparent emissions treatment.

Materials

- Hourly/annual energy profile
- Illustrative capex, opex and tariff assumptions
- Grid-factor and certificate integrity prompts
- Business-case workbook

Activity Questions

- Which option reduces demand before changing supply?
- What emissions accounting and claim conditions apply?
- What are NPV, payback and cost per tCO2e?
- Which risks and sensitivities could reverse the recommendation?

Detailed Procedure

63. Establish the energy baseline, load profile, emissions basis and business constraints.
64. Model efficiency and operational controls before new supply options.
65. Estimate onsite solar generation, self-consumption, export and annual emissions effect.
66. Evaluate process electrification including load, grid trajectory, capex and operational dependencies.
67. Assess renewable-electricity procurement and certificate quality, geography, vintage, exclusivity and retirement.
68. Calculate payback, NPV, marginal abatement cost and key sensitivity cases.
69. Score delivery, technology, regulatory, data and claim risks.
70. Recommend a staged portfolio with trigger points rather than one isolated project.
71. Prepare a one-page decision brief separating inventory effect, avoided emissions and claims.

Required Evidence

- Renewable-energy scenario model
- Option scorecard
- Sensitivity analysis
- Executive recommendation

Acceptance Check

The recommendation uses consistent assumptions, separates demand and supply levers, quantifies economics and emissions, and states certificate and claim safeguards.

Activity files: activities/08-/README.md, workbook and printable PDF/checklist.*

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Topic 04 — Designing Viable Carbon Offset and Strategy Plans

Alignment: K8 · K9 · A2 · A5 · LO4

Residual emissions · project quality · offset economics · claims · carbon-credit optimisation

Residual emissions and the role of offsets

- **Gross inventory:** Report organisational emissions and removals before external credits.
- **Reduction pathway:** Show internal and value-chain actions against a declared baseline.
- **Residual emissions:** Define what remains after feasible reduction within the target boundary.
- **Credit use:** Disclose quantity, project, vintage, ownership, retirement and claim separately.

Auditor application

- Set an offset eligibility policy
- Forecast residuals conservatively
- Procure against a clear use case
- Retire credits before claiming use

Judgement cautions

- Credits are not removals by the organisation
- Future delivery may fail
- Neutrality language can overstate scope
- Do not hide gross performance

Carbon project integrity

- **Additionality:** The claimed outcome would not occur without the project incentive under the defined test.
- **Baseline and quantification:** A credible counterfactual, monitoring method and uncertainty treatment support the tonnes.
- **Permanence and leakage:** Durability, reversal response and displaced emissions are addressed.
- **No double counting:** Issuance, ownership, transfer, host claims and retirement prevent duplicate use.

Auditor application

- Review methodology and project documents
- Check independent validation and verification
- Inspect registry status

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- Assess safeguards and grievance routes

Judgement cautions

- Registry listing is not automatic quality
- Co-benefits do not repair weak carbon accounting
- Avoidance and removal have different durability
- Buffer pools do not eliminate reversal risk

Offset economics and procurement risk

- **Unit price:** Quoted cost per credit is only the starting point.
- **Total cost:** Include due diligence, legal, registry, retirement, monitoring and internal governance.
- **Delivery risk:** Forward projects face issuance, timing, volume, policy and counterparty uncertainty.
- **Replacement exposure:** Low-quality or failed credits may require replacement at a higher future price.

Auditor application

- Model risk-adjusted cost
- Use portfolio limits
- Set conditions precedent
- Retain retirement and claim evidence

Judgement cautions

- Lowest price can carry highest risk
- Co-benefits need evidence
- Vintage must match intended use
- A long contract can lock in weak criteria

Carbon-credit portfolio optimisation

- **Purpose:** Match credit type and timing to residual emissions and claim need.
- **Quality floor:** Define non-negotiable methodology, verification, safeguards and registry conditions.
- **Diversification:** Manage project, geography, technology, delivery and counterparty concentration.
- **Governance:** Set approval, custody, retirement, accounting, disclosure and review responsibilities.

Auditor application

- Use scoring plus hard exclusions
- Stage procurement over time
- Reserve for delivery failure

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- Review criteria as science and policy evolve

Judgement cautions

- Diversification cannot legitimise weak credits
- Removal scarcity affects price
- Portfolio optimisation is not emissions optimisation
- Claims need boundary-specific wording

Integrated organisational carbon strategy

- **Inventory foundation:** A governed, repeatable inventory establishes what is measured and managed.
- **Target and pathway:** Business growth, initiatives, dependencies and residuals are reconciled transparently.
- **Capital and risk:** Investment gates balance abatement, economics, resilience, operations and data confidence.
- **Disclosure and learning:** A dashboard, annual review and claim controls keep strategy credible as conditions change.

Auditor application

- Connect every initiative to an inventory source
- Assign accountable owners and budgets
- Stress-test key assumptions
- Report gross, reduced and offset tonnes separately

Judgement cautions

- A strategy is not a list of projects
- Targets require governance and recalculation
- Do not combine avoided and inventory emissions
- Update claims when evidence changes

Activity 9 — Evaluate Carbon Offset Projects and Economics

Alignment: K8 · K9 · A2 · LO4 · Duration: 50 minutes

Objective: Assess offset quality, delivery and financial implications before procurement.

Scenario

EcoForge receives proposals for a tropical forest avoidance project, a landfill-methane project and an engineered carbon-removal contract. Prices, vintages, delivery dates, durability, buffer provisions, host-country authorisation, verification status and co-benefits differ. Marketing wants to claim carbon neutrality next quarter.

Materials

- Three fictional project dossiers

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- Offset integrity and claims criteria
- Price and transaction-cost assumptions
- Due-diligence workbook

Activity Questions

- What evidence supports additionality and robust baselines?
- How are permanence, leakage and reversal addressed?
- Could double counting or delivery failure occur?
- What is the risk-adjusted total cost per retired tonne?

Detailed Procedure

72. Confirm EcoForge's intended use, residual-emissions boundary, quantity, timing and claim objective.
73. Review each project's methodology, baseline, additionality rationale and monitoring evidence.
74. Assess permanence or durability, reversal response, leakage and uncertainty deductions.
75. Check independent validation/verification, registry serialisation, ownership, retirement and double-counting controls.
76. Evaluate safeguards, stakeholder impacts and credible co-benefits without treating them as substitutes for climate integrity.
77. Model credit price, transaction cost, due diligence, delivery risk and replacement exposure.
78. Score quality and risk, record red flags and specify conditions precedent to purchase.
79. Recommend purchase, pilot, defer or reject for each project and explain why.
80. Draft claim language that clearly states inventory boundary, reduction actions, credit quantity, vintage and retirement.

Required Evidence

- Due-diligence register
- Quality scorecard
- Risk-adjusted cost model
- Procurement and claims recommendation

Acceptance Check

The decision is evidence-based, includes quality and financial risk, requires retirement proof, and does not use credits to obscure gross organisational emissions.

Activity files: activities/09-/README.md, workbook and printable PDF/checklist.*

Activity 10 — Integrate the Organisational Carbon Strategy

Alignment: K2 · K7 · K8 · K9 · A2 · A5 · LO4 · Duration: 30 minutes

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Objective: Combine inventory governance, reduction pathways, renewable energy and high-integrity offsets into an executable strategy.

Scenario

EcoForge expects production to grow 35% by 2030. The board will fund projects only if cash flow and operational resilience remain acceptable. Customers demand supplier-specific reductions, the lender tests performance annually, and Marketing wants a simple climate claim. The strategy must show gross emissions, reductions, residuals and any credit use transparently.

Materials

- Outputs from Activities 1-9
- 2030 growth and budget assumptions
- Initiative cost/abatement portfolio
- Board decision and challenge cards

Activity Questions

- What target and boundary are credible and decision-useful?
- Which sequence minimises emissions and strategic risk?
- How are residual emissions and credits governed?
- Which dashboard indicators trigger review or recalculation?

Detailed Procedure

81. Consolidate the verified baseline, boundaries, significant sources, uncertainty and growth assumptions.
82. Define target type, base year, target year, boundary, metric and treatment of structural changes.
83. Build a business-as-usual pathway and a reduction pathway using quantified initiatives from Activities 6-8.
84. Sequence avoid, reduce, substitute and renewable-energy actions by dependency, cost, risk and abatement.
85. Forecast residual emissions and define when offsets may be considered and what quality rules apply.
86. Set a carbon-credit budget, procurement schedule, approval authority, retirement evidence and claim controls.
87. Define governance, funding gates, supplier/customer engagement and annual inventory/review cadence.
88. Create a board dashboard covering gross emissions, intensity, reductions, data quality, risk, capex and retired credits.
89. Stress-test the strategy against high growth, grid-factor change, project delay and credit-price shock.
90. Present the strategy and respond to Finance, Operations, Procurement, Sustainability and Marketing challenges.

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Required Evidence

- 2030 pathway model
- Prioritised initiative portfolio
- Offset and credit policy
- Board dashboard and decision memo

Acceptance Check

The strategy reconciles baseline, pathway and residuals; reduction precedes offsetting; financial and delivery risks are explicit; targets, credits and claims have accountable governance.

Activity files: activities/10-/README.md, workbook and printable PDF/checklist.*

Assessment Preparation

Written Assessment (WA-SAQ): 9 open-ended questions covering K1-K9, 1 hour, open book.

Case Study (CS): one coherent scenario with 4 open-ended questions covering A1-A8, 1 hour, open book.

Competency requires a Competent result for every assessed K and A criterion and at least 75% attendance.

Sources and Further Reading

- Tertiary course page: https://www.tertiarycourses.com.sg/wsq-iso-14064-1-2018-greenhouse-gas-ghg-emissions-and-removals.html
- ISO 14064-1:2018 official catalogue: https://www.iso.org/standard/66453.html
- ANSI overview of ISO 14064-1:2018: https://blog.ansi.org/ansi/iso-14064-1-2018-greenhouse-gas-emissions/
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- ISO 14064-1:2018 public sample: https://cdn.standards.iteh.ai/samples/66453/2868e171e2654980b6db87375ad1a20f/ISO-14064-1-2018.pdf

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- Environment Ireland ISO 14064 presentation: https://www.environmentireland.ie/presentations_2021/Liam_Regan.pdf
- GHG Protocol Corporate Standard: https://ghgprotocol.org/corporate-standard
- IPCC Sixth Assessment Report: https://www.ipcc.ch/assessment-report/ar6/

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